

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2022-23

Paper: GLB-101 — Elementary Physical and Structural Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 055637

*Instructions: Answer **five** questions, selecting at least **two** each from Sections A and B, and Section-C which is compulsory.*

SECTION — A

Question 1. With a neat block diagram of a fault, define different elements of it. Describe different types of fault based on the rake of the net slip and attitude of fault relative to the attitude of the adjacent rocks with neat diagrams. [14]

Question 2. Describe the difference between the conformable strata and unconformity in a sequence of strata. Discuss the different types of unconformities along with their origin with neat diagrams. [14]

Question 3. Discuss how physical processes control weathering and erosion of a rock. [14]

Question 4. Discuss the discontinuities in the Earth's interior. Discuss with diagram the passages of P and S waves inside the Earth. [14]

SECTION — B

Question 5. Write short notes on the following: [2 × 7 = 14]

- (a) Discordant igneous structures.
- (b) Strike and dip and right-hand rule of recording of strike.

Question 6. Discuss the following: [2 × 7 = 14]

- (a) Outlier and inlier with neat diagram.
- (b) Types of sedimentary rocks.

SECTION — C

Question 7. Briefly discuss the following: [4 × 3½ = 14]

- (a) Strike and dip joints.
- (b) Basic features of contour lines.
- (c) Origin of Earth's second atmosphere.

(d) Causes of earthquake.

Question 8. Write about each of the following in 2 to 3 sentences. Give suitable diagrams wherever necessary: [14 × 1 = 14]

(a) Contour map of conical hill

(b) Inflection point

(c) Axial plane

(d) Antiformal syncline

(e) Overturned antiform

(f) Outcrop and incrop

(g) Relationship between horizontal bed and contour lines

(h) The biggest mass extinction event in the Earth's history was during the _____.

(i) Nebular hypothesis was proposed by _____.

(j) Stratovolcano

(k) Index fossil with a suitable example

(l) Name two metamorphic rocks

(m) Half life

(n) Earth's magnetic field is generated by _____.